



GAMEWATCH

PLAY • MANAGE • CONNECT

The All-in-One PS Management System
for Staff, Players & Customers



PLAY
FOR PLAYERS



MANAGE
FOR STAFF



CONNECT
FOR CUSTOMERS

Master Product Requirements Document

PRD v1.0 — Product Blueprint

Core principle: Automate what the camera can reliably know,
let humans record what only humans can know,
and give the owner the complete picture.

Document Overview

Product	GameWatch
Purpose	Game-lounge operating, monitoring, payment, staff, customer, tournament, and business-intelligence platform
Primary market	Game/FIFA/FC lounges, initially Ethiopia, designed for international expansion
Core principle	Automate what the camera can reliably know, let humans record what only humans can know, and give the owner the complete picture.

1. Product Vision

GameWatch starts with one problem:

Game-lounge owners have TVs, customers, clerks, money and cameras, but very little automated understanding of what is actually happening.

GameWatch connects those pieces.

```
Cameras
  ↓
TV Detection
  ↓
Scoreboard Reader
  ↓
Match Verification
  ↓
Game Counter
  ↓
Sessions / Payments / Staff
  ↓
Owner Dashboard
  ↓
Business Intelligence
```

Then the customer side builds on top:

```
Lounge
  ↓
Customer
  ↓
Available TVs
  ↓
Happy Hour
  ↓
Promotions
  ↓
Tournaments
  ↓
Community
```

2. Three Main Users

OWNER

The business owner has full access.

Can:

- monitor cameras
- manage TVs
- manage prices
- manage clerks
- monitor shifts
- see revenue
- see payments
- see adjustments
- see complaints
- see maintenance
- see historical records
- see analytics
- see weekly/monthly reports
- create Happy Hours
- create promotions
- create tournaments
- manage customers/community settings
- manage subscriptions
- manage payment methods
- manage cameras
- manage lounge settings

3. CLERK

The clerk is an **operations user**, not a business-management user.

Can:

- log in with assigned lounge code + personal PIN
- use authorized device
- start shift
- manage today's sessions
- add/correct games
- record payments
- upload payment evidence
- report problems
- report customer complaints
- report damaged equipment
- upload photos
- handle major incidents
- end shift
- record cash/deposit information where required

Cannot normally see:

- historical revenue
- weekly business reports
- monthly reports
- owner analytics

- other clerks' private information
- subscription
- lounge settings
- business intelligence
- staff management

4. CUSTOMER

Customers get a separate experience.

They can:

- discover/follow a lounge
- see lounge information
- see available TVs
- see estimated waiting time
- join a waiting list
- see Happy Hour
- see promotions
- see tournaments
- join tournaments
- see tournament schedules
- see their match time
- see tournament brackets
- see results
- receive lounge notifications

5. Login / Account Flow

First screen

GAMEWATCH

Smart game-lounge management

[OWNER]
[CLERK]
[CUSTOMER]

Owner

Owner creates or logs into a lounge.

Clerk

Clerk receives credentials from owner.

Customer

Customer can use the customer experience and connect to a particular lounge.

6. Owner Account Creation

CREATE LOUNGE

Lounge name
Owner name
Phone/email
Password

Country
Currency

[Continue]

Then:

Lounge Setup

Number of TVs
Payment methods
Opening hours
Default game price
Camera setup

The owner can continue configuring the lounge later.

7. Clerk Security Model

The owner creates a clerk.

ADD CLERK

Name
Phone

Generate access code
Create personal PIN

Example:

Lounge Code: GW-4827
Clerk PIN: 5831

The code is **not a permanent reusable public invitation**.

The clerk is associated with:

Lounge
+
Clerk identity
+
Authorized device

The system records active devices.

If the same clerk suddenly appears from another device:

New device detected.

Owner can investigate/revoke access.

The system should maintain an **activity/login audit trail**.

8. Clerk Daily Workspace

The clerk should not see a giant business dashboard.

Her interface is intentionally simple.

TODAY

[LIVE TVs]

[MANAGE GAMES]

[PAYMENTS]

[REPORT PROBLEM]

[MY SHIFT]

9. Shift System

When the clerk arrives:

Good morning, Hana.

Your previous shift:
Yesterday 17:02 – 23:14

Today's workspace is ready.

[START SHIFT]

When she starts:

SHIFT STARTED

08:57 AM

Games: 0
Sessions: 0
Adjustments: 0
Problems: 0

GameWatch records:

- clerk
- date/business day
- start time
- device
- shift status

10. Business Day

This is important because a lounge may operate past midnight.

The system should support a **business day**, rather than blindly treating midnight as the end of the working day.

For example:

October 2 business day
10:00 AM → 2:00 AM

The next business day starts according to the lounge's configured schedule.

The clerk's workspace becomes a fresh daily workspace.

11. Blank Next-Day Workspace

The clerk starts the next day with:

```
OCTOBER 3

Games: 0
Sessions: 0
Adjustments: 0
Problems: 0

[ START SHIFT ]
```

Yesterday's data is **not deleted**.

It simply isn't part of the clerk's normal daily workspace.

The owner retains the historical records.

12. End Shift

```
END SHIFT

Started:
08:57

Games handled:
48

Adjustments:
3

Problems:
2

Cash:
800 ETB

Telebirr:
400 ETB

Total:
1,200 ETB

[ END SHIFT ]
```

GameWatch records the exact end time and the state of the lounge/work performed.

13. Owner Dashboard

The owner's home screen is the main command center.

```
GAMEWATCH

TODAY

Revenue      Games
1,425 ETB    57

Active TVs   Clerks
5 / 8        2
```

Then:

Live Lounge

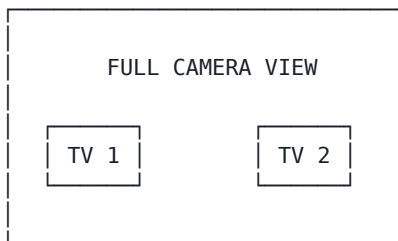
- full camera feeds
- detected TVs
- cropped TV regions
- current match state
- clock
- detection confidence
- TV status

14. Full Camera View

Owner should be able to see the complete camera image.

Example:

CAMERA 1



Then switch to:

TV Crop View

TV 1

[cropped scoreboard/TV area]

Clock: 37:42
Status: ACTIVE
Confidence: 94%

This gives the owner transparency into **what the system actually sees**.

15. TV Management

Owner can add TVs.

ADD TV

TV name
Location
Game
Price per game

Connected camera
Detected region

[SAVE]

Each TV has its own:

- name
- location
- price
- camera
- detection region
- game
- status
- utilization
- revenue
- maintenance history

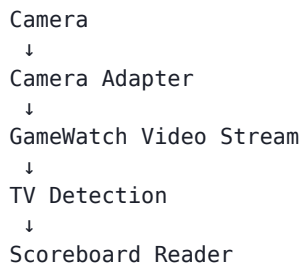
16. Camera Management

GameWatch should be camera-agnostic.

Supported input categories:

- phone camera
- PC/webcam
- USB camera
- security camera
- CCTV/IP camera
- compatible network video stream

The architecture should normalize these into one internal video interface.



One camera may potentially monitor multiple TV regions.

17. Camera Health

GameWatch monitors detection quality.

Potential statuses:

- Healthy
- Poor detection
- Disconnected
- Obstructed
- Too blurry
- Too dark

Example:

Camera 3

TV 4 detection confidence has been unusually low.

18. AI / Antigravity Layer

Antigravity should be an **analysis layer**, not the authority for financial decisions.

Core GameWatch decides:

"I detected this game."

AI can analyze:

"Why is detection failing?"

Example:

TV 4

Detection confidence:
normally 92–95%

Today:
61–70%

AI analysis:
Camera position appears to have changed
or the scoreboard region is partially obscured.

This separation is important.

AI explains. Deterministic GameWatch rules decide.

19. Game Detection

Core pipeline:

```
Camera
↓
TV Region
↓
Scoreboard Crop
↓
Preprocessing
↓
Scoreboard Reader
↓
Clock
↓
Confidence
↓
Time-Aware Brain
↓
Match Verifier
↓
Completed Game
```

The existing project has already reached the scoreboard-reader stage, with the next integration being the time-aware brain and match verifier.

20. Game Counter

GameWatch records:

TV 3

Game completed
Price: 25 ETB
Time: 18:43
Detection confidence: 94%

The game automatically becomes part of the lounge's records.

21. Manual Game Entry

If the clerk manually adds a game:

ADD GAME

TV
Game
Price

Why was this added manually?

- ☐ GameWatch failed to detect it
- ☐ Camera unavailable
- ☐ Customer arrangement
- ☐ Loyal customer/package
- ☐ Other

If needed:

Explain what happened.

This creates intelligence about **why automatic detection isn't matching reality**.

22. Manual Discount / Adjustment

A clerk cannot simply erase revenue.

Instead:

ADJUST TRANSACTION

Original:
25 ETB

Reason:

- ☐ Customer did not finish
- ☐ Technical problem
- ☐ Detection error
- ☐ Loyal customer
- ☐ Package
- ☐ Other

Explanation:

[SUBMIT]

Every adjustment is audited.

23. Customer Discount Example

20 games:

20 × 25 = 500 ETB

Customer receives a package/loyalty discount:

Final = 400 ETB

GameWatch records:

Normal value: 500 ETB
Discount: 100 ETB
Final payment: 400 ETB

Reason:
Loyal customer/package

Owner can analyze these discounts later.

24. Payments

Owner can configure payment methods.

Example Ethiopia setup:

- Cash
- Telebirr
- CBE

Each lounge can configure its own supported payment methods.

25. Payment Evidence

For digital payment:

Customer pays.

Clerk records:

Payment method:
Telebirr

Amount:
400 ETB

Payment proof:
[Upload photo]

[SAVE]

The payment screenshot/photo becomes attached to the transaction.

Same concept for CBE.

26. Cash Settlement

At the end of the shift/day:

EXPECTED CASH

800 ETB

ACTUAL CASH

780 ETB

Difference:
-20 ETB

Reason:

[SUBMIT]

Or the owner can require a deposit.

DEPOSIT

Amount
Bank
Receipt photo

[SUBMIT]

The owner sees the record.

27. Problem Reporting

Clerk:

REPORT PROBLEM

Type:

- ☐ Controller damaged
- ☐ Controller missing
- ☐ TV problem
- ☐ Console problem
- ☐ Internet problem
- ☐ Payment problem
- ☐ Camera problem
- ☐ Customer complaint
- ☐ Other

TV:
TV 3

Description:

Photo:
[Add]

Priority:
Normal / Important / Major

[SUBMIT]

28. Major Incidents

For major issues:

Call Owner

while also recording the incident.

Owner receives:

MAJOR ISSUE

TV 3
Controller damaged

Reported by:
Hana

18:43

Photo attached

[INVESTIGATE]

Status:

OPEN → INVESTIGATING → FIXED → CLOSED

29. Maintenance Intelligence

GameWatch keeps the history.

Example:

TV 3

Controller reports:
Sept 21
Sept 25
Sept 29

The owner can then see:

Repeated controller-related problems have been recorded for TV 3.

This can feed the intelligence system.

30. TV Business Analytics

Every TV gets its own business profile.

TV 1

Games: 24
Revenue: 600 ETB
Utilization: 41%

Detection:
Good

Problems:
4

Customer reports:
3

The owner can investigate underperforming TVs.

31. Low-Utilization Analysis

GameWatch should not simply declare:

"TV 1 is bad."

Instead:

Observed

TV 1 had lower utilization than the lounge average.

Related recorded data

- lower game count
- controller complaints
- availability periods
- camera problems
- location
- customer reports

Possible explanations

Controller issues may be contributing.

TV location may be affecting usage.

Customers may prefer other TVs.

These are **analysis/hypotheses**, not fabricated facts.

32. Happy Hour

Owner can create:

HAPPY HOUR

Days:
Monday–Thursday

Time:
2 PM – 5 PM

Normal price:
25 ETB

Happy Hour:
15 ETB

[ACTIVATE]

GameWatch can also identify consistently low-utilization periods and suggest potential Happy Hour windows.

The owner makes the final decision.

33. Customer Happy Hour

Customer sees:

HAPPY HOUR

2 PM – 5 PM

15 ETB / GAME

Ends in 01:24

[VIEW LOUNGE]

34. Customer TV Availability

LIVE AVAILABILITY

TV 1 [Playing]

TV 2 [Available]

TV 3 [Available]

TV 4 [Ending soon]

Customer can avoid unnecessary waiting.

35. Waiting List

If all TVs are busy:

ALL TVs BUSY

Estimated wait:

18 minutes

You are:

#3 in line

[JOIN WAITING LIST]

When a TV becomes available:

TV 4 is available.

36. Customer Lounge Page

Each participating lounge can have:

Champion Game Lounge

8 TVs

FIFA / FC

3 TVs available

Happy Hour

Tournament

Promotions

[Follow Lounge]

[View TVs]

[Tournaments]

37. Lounge Promotions

Owner can publish announcements.

Examples:

New TVs have arrived

Cold drinks available

More space available

New PS5s installed

Tournament this Saturday

Happy Hour 2-5 PM

Owner can attach an image.

Customers following the lounge receive the promotion.

38. Tournament Creation

Owner:

CREATE TOURNAMENT

Name

Date

Start time

Game

Maximum players

Entry fee

Prize

Format

[CREATE]

Example:

32 players × 200 ETB = 6,400 ETB gross entry revenue

Winner prize = 1,000 ETB

Other costs, if any, are separately recorded before calculating net tournament revenue.

39. Tournament Registration

Customer:

FRIDAY NIGHT CHAMPIONSHIP

32 Players

Entry:

200 ETB

Winner:

1,000 ETB

16 / 32 spots filled

[JOIN]

40. Tournament Bracket

GameWatch automatically maintains:

```
Round of 32
  ↓
Round of 16
  ↓
Quarter Finals
  ↓
Semi Finals
  ↓
FINAL
  ↓
WINNER
```

Every match result is recorded.

41. Customer Tournament Experience

Customer sees:

MY TOURNAMENT

Quarter Final

Opponent:
Messay

TV:
TV 4

Estimated time:
7:40 PM

[VIEW BRACKET]

After the match:

You won 3-1.

or:

You lost 3-2.

The result updates the bracket.

42. Tournament Waiting

The customer doesn't need to sit beside the TV all day.

GameWatch can show:

Your match is estimated for 8:15 PM.

Then:

Your match is coming up.

This makes the tournament experience much more organized.

43. Tournament Business Dashboard

Owner sees:

FRIDAY NIGHT CHAMPIONSHIP

Players:
32 / 32

Entry revenue:
6,400 ETB

Prize:
1,000 ETB

Matches:
24 / 31

Current round:
Quarter Final

44. Weekly Business Report

Owner:

WEEKLY REPORT

Games:
343

Revenue:
8,575 ETB

Average games/day:
49

Average revenue/day:
1,225 ETB

Then:

TV performance

Peak hours

Payment breakdown

Discounts

Adjustments

Clerk activity

Maintenance

Detection failures

Tournament performance

45. Business Intelligence

Eventually:

GameWatch Intelligence

THIS WEEK

Revenue up 12%

Games up 8%

INSIGHTS

! TV 1 utilization is low.

! TV 3 had repeated controller reports.

Camera 3 had poor detection
confidence several times.

6 PM–9 PM is the busiest period.

Thursday 2 PM–5 PM has
consistently low utilization.

Again, the system distinguishes:

DATA → ANALYSIS → POSSIBLE EXPLANATION

46. Customer Packages

The loyalty/discount system can eventually support:

20 GAME PACKAGE

Normal:
500 ETB

Package:
450 ETB

Games remaining:
14

This avoids clerks manually inventing discounts every time.

47. Internationalization

Country is a lounge setting.

Country
Currency
Language
Payment methods

The price system must never hard-code ETB.

For Ethiopia:

25 ETB

Another country can configure its own currency and pricing.

48. Settings

Lounge

- Name
- Logo
- Country
- Currency
- Address
- Opening hours

TVs

- Add
- Remove
- Rename
- Price
- Game
- Location

Cameras

- Add
- Remove
- Camera type
- Connection
- TV regions
- Health

Payments

- Add payment method
- QR code
- Bank information

Staff

- Add clerk
- Disable clerk
- Generate credentials
- Devices

- Activity

Notifications

- Camera problems
- Major incidents
- Revenue discrepancies
- Device login alerts

Security

- Password
- PIN
- Active devices
- Login history

Subscription

- Current plan
- Trial
- Renewal
- Upgrade

49. Contact GameWatch

CONTACT US

Phone
Email

[Contact Support]

[Report Technical Problem]

Technical reports can include screenshots/photos and relevant system information.

50. Advertising

GameWatch has an advertising area.

Example:

ADVERTISEMENT

Local gaming store
Controllers • Consoles • Accessories

Then:

Want your business advertised here?

Contact GameWatch.

Advertising becomes an additional GameWatch revenue stream.

51. Subscription Model

FREE

1 TV

Includes:

- Basic dashboard
- Basic detection
- Basic game tracking
- Basic history

Does **not** include:

- Clerk access
- Multiple TVs
- Advanced analytics
- Advanced camera management
- Tournament system
- Advanced customer features

52. PLUS

For growing lounges.

Includes:

- Multiple TVs
- Multiple cameras
- Clerk accounts
- Shift management
- Payments
- Payment evidence
- Daily/weekly reports
- Problems/maintenance
- Customer lounge
- Happy Hour
- Promotions
- Basic analytics

53. PRO

For larger/serious lounges.

Everything in Plus, plus:

- Advanced business intelligence
- AI analysis
- Advanced camera monitoring
- Tournament system
- Customer community
- Tournament brackets
- Advanced reports
- TV utilization intelligence

- Maintenance intelligence
- Advanced customer engagement
- Advanced promotions

54. Free Trial

30-Day Free Trial

User enters payment information during signup.

The system clearly tells them:

30-day free trial. You will be charged after the trial according to the selected plan unless you cancel.

Then subscription choices:

3 Months

Paid plan

6 Months

Paid plan

1 Year

Paid plan

Pricing is configured separately.

55. Core Database Model

The backend should be designed around these major objects:

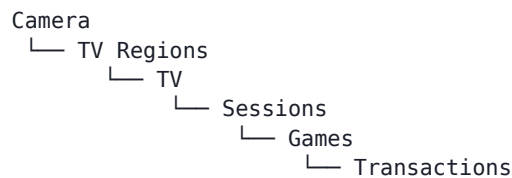
User
Lounge
Subscription
Clerk
Device
Camera
TV
TVRegion
Game
Session
Match
Transaction
Payment
PaymentProof
Adjustment
Shift
ProblemReport
MaintenanceRecord
Customer
LoungeFollower
HappyHour
Promotion
Tournament
TournamentPlayer
TournamentMatch
TournamentResult
Notification
AuditLog

This gives us a foundation without forcing everything into one giant table.

56. Most Important Relationships



And:



57. Audit Log

This should be everywhere money or permissions are involved.

GameWatch records:

- who did it
- what happened
- when
- from which device
- previous value
- new value
- reason

Example:

18:42

Hana

Transaction #1042

25 ETB → 0 ETB

Reason:
Technical problem

Device:
Samsung A03

This is extremely important for business trust.

58. Product Rules We Should Not Break

Rule 1

AI does not decide financial truth.

Rule 2

Clerk adjustments require reasons.

Rule 3

Historical records are never destroyed merely because a new day starts.

Rule 4

Owner has full visibility.

Rule 5

Clerk sees only what she needs to operate.

Rule 6

Customer cannot access private lounge business information.

Rule 7

Camera failure must be visible rather than silently producing fake data.

Rule 8

GameWatch should never pretend it knows something it did not observe.

Rule 9

Currency and payment methods are configurable by lounge.

Rule 10

Every important financial/security action is auditable.

59. Development Roadmap

This is where we put our actual development.

PHASE 1 — Detection Engine

We are here.

```
Camera
↓
TV detection
↓
Scoreboard preprocessing
↓
Scoreboard reader
↓
Time-aware brain
↓
Match verifier
↓
Game counter
```

The scoreboard reader has already been implemented and tested against the current captured examples; the project log records successful readings of 00:51, 01:21 and 03:15 with confidence

values in roughly the 89–94% range.

PHASE 2 — Core Backend

Build:

- database
- lounge
- TVs
- cameras
- sessions
- games
- transactions
- adjustments
- audit logs

PHASE 3 — Owner UI

Build:

- dashboard
- live camera
- TV management
- camera management
- sessions
- revenue
- reports
- staff
- settings

PHASE 4 — Clerk UI

Build:

- login
- device authorization
- daily workspace
- start shift
- games
- payments
- evidence
- problems
- adjustments
- end shift

PHASE 5 — Customer

Build:

- lounge page
- available TVs
- Happy Hour
- promotions
- notifications
- follow lounge

PHASE 6 — Tournaments

Build:

- tournament creation
- registration
- payment
- bracket
- matches
- results
- notifications
- winner/prize records

PHASE 7 — Intelligence

Build:

- TV utilization
- low-use analysis
- peak hours
- maintenance patterns
- detection-quality analysis
- revenue patterns
- Happy Hour suggestions
- Antigravity analysis layer

PHASE 8 — Monetization

Build:

- free plan
- 30-day trial
- Plus
- Pro
- 3-month
- 6-month
- 1-year subscriptions
- advertising

60. The Final GameWatch Product

If someone asks us:

"What is GameWatch?"

The answer is:

GameWatch is a smart operating platform for game lounges that uses existing cameras and computer vision to automatically understand game activity, while giving owners tools for staff management, payments, revenue tracking, maintenance, promotions, Happy Hours, tournaments, customer engagement and business intelligence.

And the three experiences are crystal clear:

Owner

"Show me my business."

Clerk

"Help me run today."

Customer

"Help me find a game, know when to play, and have more fun."

GameWatch Intelligence

"Help everyone understand what is happening."

This is the **GameWatch PRD v1.0**. When we build something, we check it against this blueprint instead of randomly changing the product architecture.